

283. Move Zeroes

Given an integer array `nums`, move all 0's to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

Input: `nums = [0,1,0,3,12]`

Output: `[1,3,12,0,0]`

Example 2:

Input: `nums = [0]`

Output: `[0]`

The screenshot shows a LeetCode problem page for "283. Move Zeroes". The left sidebar contains the problem description, examples, and constraints. The right sidebar shows the code editor with a Python solution and the test results section.

Problem Description: Given an integer array `nums`, move all 0's to the end of it while maintaining the relative order of the non-zero elements. **Note** that you must do this in-place without making a copy of the array.

Example 1:
Input: `nums = [0,1,0,3,12]`
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Example 2:
Input: `nums = [0]`
Output: `[0]`

Constraints:

- $1 \leq \text{nums.length} \leq 10^4$
- $-2^{31} \leq \text{nums}[i] \leq 2^{31} - 1$

Follow up: Could you minimize the total number of operations done?

Code:

```
class Solution:
    def moveZeroes(self, nums: List[int]) -> None:
        """
        Do not return anything, modify nums in-place instead.
        """
        nonzero = 0
        for current in range(len(nums)):
            if nums[current] != 0:
                nums[nonzero], nums[current] = nums[current], nums[nonzero]
                nonzero = nonzero + 1
```

Test Results: Accepted Runtime: 0 ms

Case 1: Input: `nums = [0,1,0,3,12]` Output: `[1,3,12,0,0]`

Case 2: Input: `nums = [0]` Output: `[0]`